

Write a Prolog Program to Implement a Simple Database with NAME and DOB

Aim

To develop a Prolog program that stores and retrieves information from a simple database containing a person's **Name** and **Date of Birth**.

Objective

- To understand facts in Prolog.
- To create a simple knowledge base.
- To retrieve information using queries.
- To demonstrate logical reasoning using facts.

Theory

A **database in Prolog** is represented using facts. Facts describe relationships or properties of objects. The Prolog interpreter searches the knowledge base and returns matching results for a given query. In this experiment, each person's name and date of birth are stored as facts, and queries are used to retrieve the required information.

Algorithm

Step 1: Define facts containing names and dates of birth.

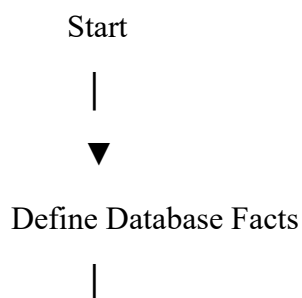
Step 2: Save the facts in the Prolog knowledge base.

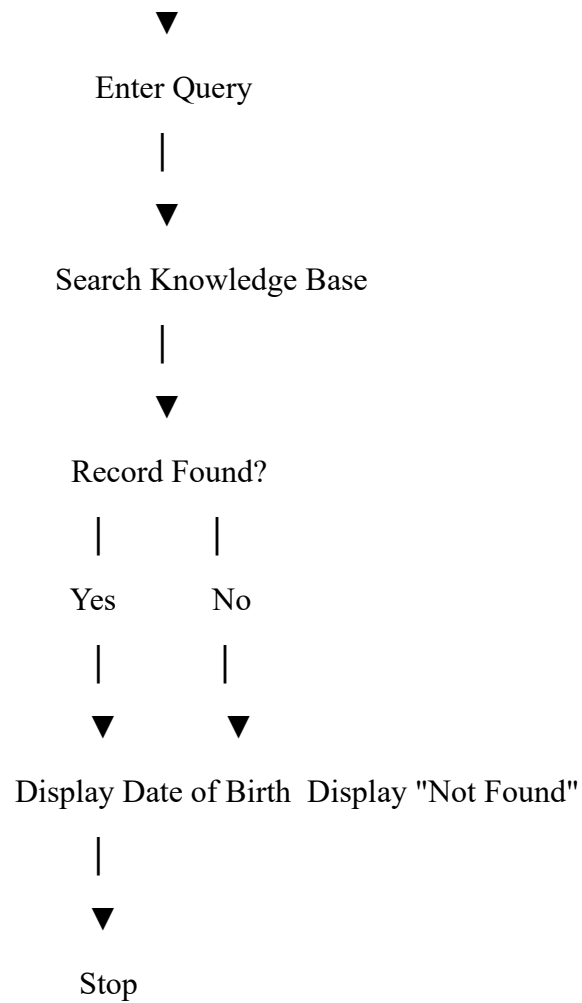
Step 3: Execute a query for a specific person.

Step 4: Search the knowledge base.

Step 5: Display the corresponding date of birth.

Flowchart





Prolog Program

dob(ram,'12-05-2000').

dob(sita,'25-08-2001').

dob(arun,'15-10-2002').

dob(priya,'20-01-2003').

Sample Output

?- dob(ram,D).

D = '12-05-2000'.

?- dob(priya,D).

D = '20-01-2003'.

?- dob(kumar,D).

false.

Result

The Prolog program successfully stored and retrieved records from the database using logical queries.

Conclusion

This experiment demonstrates how Prolog represents knowledge using facts and retrieves information through logical inference. It forms the foundation for building expert systems and knowledge-based applications in Artificial Intelligence.

```
*
9  # Display all records
10 print("Date of Birth Records:")
11 for name, date in dob.items():
12     print(f"{name} : {date}")
13
14 # Search for a person's DOB
15 person = input("\nEnter name to find DOB: ").lower()
16
17 if person in dob:
18     print(f"DOB of {person} is {dob[person]}")
19 else:
20     print("Record not found")
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\Varshini M\OneDrive\Desktop\Here!!!\ARTIFICIAL INTELLIGENCE\VSC EXPS> & "C:\Users\Varshini M\AppData\Local\Programs\Python\Python311\python.exe" "C:\Users\Varshini M\OneDrive\Desktop\Here!!!\ARTIFICIAL INTELLIGENCE\VSC EXPS\EXP1.py"
15
PS C:\Users\Varshini M\OneDrive\Desktop\Here!!!\ARTIFICIAL INTELLIGENCE\VSC EXPS> & "C:\Users\Varshini M\AppData\Local\Programs\Python\Python311\python.exe" "C:\Users\Varshini M\OneDrive\Desktop\Here!!!\ARTIFICIAL INTELLIGENCE\VSC EXPS\EXP1.py"
Date of Birth Records:
nam : 12-05-2000
sita : 25-08-2001
arun : 15-10-2002
priya : 20-01-2003

Enter name to find DOB: sita
DOB of sita is 25-08-2001
```